Solid State Nuclear Magnetic Resonance 137 (2025) 102006

Contents lists available at ScienceDirect

Solid State Nuclear Magnetic Resonance

journal homepage: www.elsevier.com/locate/ssnmr

Low-power ^{14}N decoupling at fast MAS of 70 kHz

Ekta Nehra [a], Vipin Agarwal [b,*], Yusuke Nishiyama [a,**]

[a] JEOL Ltd., Akishima, Tokyo, 196-8558, Japan
[b] Tata Institute of Fundamental Research Hyderabad, Hyderabad, Telangana, 500046, India

ARTICLE INFO

Handling Editor: Prof D Bryce

ABSTRACT

Higher isotropic resolution in solids is essential for accurate assignment of the chemical shifts in multisite chemical and biological systems; consequently, the pursuit of solution-like resolution in solid samples remains an ongoing challenge. MAS provides the effective averaging of the anisotropic interactions to the first-order while the higher order interactions, such as residual dipolar splitting (RDS), and isotropic interactions like J coupling remain. With the emergence of faster MAS, RDS and J couplings become significant constraints in the resolution and sensitivity of isotropic ^{1}H peaks bonded to ^{14}N. However, the dominant quadrupolar coupling hampers the decoupling of ^{1}H-^{14}N, since the achievable ^{14}N RF-field strength is much smaller than the size of the quadrupolar coupling. Therefore, ^{14}N edited spectroscopy is typically performed in the absence of ^{14}N decoupling, leading to broader ^{1}H linewidth and reduced sensitivity. In this context, we propose the continuous-wave (CW) ^{14}N decoupling of ^{14}N-^{1}H spin pair under ^{1}H detection at a fast MAS of 70 kHz. We experimentally show that the on-resonance low-power ^{14}N CW irradiation at fast MAS yields the narrower linewidth. Utilizing the quadrupolar jolting frame description, a qualitative analysis of the optimum decoupling effect is provided.

1. Introduction

The accuracy of the structure, dynamics, and ligand binding studies in solid-state NMR (ssNMR) spectroscopy heavily relies on the availability of the highly resolved isotropic spectrum for various NMR active nuclei [1,2]. In the case of the spin-1/2 nuclei, poor resolution that stem from the overlapping resonances in multisite systems, smaller chemical shift dispersion (e.g., ^{1}H) and inherent 2nd rank anisotropic broadenings are observed in rigid solids. As a result, recent advancements in ssNMR primarily focus on achieving high isotropic resolution [3]. With the advent of faster magic-angle spinning (MAS), it is possible to average out the 2nd rank tensorial interactions to the first-order in ssNMR [4–7]. However, MAS falls short in complete averaging of the higher-order anisotropic broadenings, such as residual ^{1}H–^{1}H dipolar couplings, and residual dipolar splitting (RDS) in the spin-1/2 nuclei in spatial proximity to quadrupolar nuclei [8,9] along with isotropic J coupling. While residual ^{1}H–^{1}H broadenings can be reduced by faster MAS, anisotropic part of RDS can potentially be completely averaged out by the DAS [10] and DOR [11]. Nevertheless, the line broadening/splitting of spin-1/2 coupled to ^{14}N spins due to RDS and isotropic J couplings remain under MAS [12–14]. RDS arises due to the second-order cross terms between the quadrupolar and dipolar interactions and results in asymmetric broadening of the peaks of spin-1/2 nuclei directly connected to or in close vicinity of quadrupolar nuclei. RDS along with J coupling, has long been a source of the structural constraints and acted as a medium of polarization transfer in heteronuclear correlation [13,14]. Particularly, the RDS between the ^{13}C and ^{14}N spin pair has received considerable attention in the existing literature owing to their prevalence in many biologically important compounds [15,16]. At faster MAS, from the perspective of resolution, RDS and J couplings are now becoming a concern in ^{1}H NMR, hence require heteronuclear spin-spin decoupling. Heteronuclear decoupling of quadrupolar nuclei connected to spin-1/2 nucleus has remained a challenge in ssNMR due to the large magnitude (in the MHz range) and anisotropic nature of quadrupolar coupling. The currently available RF-field strengths do not uniformly alter the ^{14}N spin-states across all crystallite orientations, thus, the application of the heteronuclear decoupling is very limited in the case of quadrupolar nuclei [17–20].

In the past, several methods based on the adiabatic decoupling have been implemented to eliminate the ^{14}N-X spin-spin coupling at slow

This article is part of a special issue entitled: Advancements in SSNMR published in Solid State Nuclear Magnetic Resonance.

* Corresponding author.
** Corresponding author.
E-mail addresses: vipin@tifrh.res.in (V. Agarwal), yunishiy@jeol.co.jp (Y. Nishiyama).

https://doi.org/10.1016/j.ssnmr.2025.102006
Received 19 February 2025; Received in revised form 9 April 2025; Accepted 12 May 2025
Available online 13 May 2025
0926-2040/© 2025 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

MAS. Orr et al. demonstrated the decoupling of residual dipolar coupling between the ^{13}C and ^{14}N spin pair using the train of pulses with a varying rate of pulsing [19]. This decoupling utilizes adiabatic passage; therefore, it is also reported that the decoupling efficiency is nearly independent of ^{14}N offset. Subsequently, Emsley et al. [20] showed that the asynchronous application of pulses to ^{14}N in the presence of the ^{1}H–^{1}H homonuclear dipolar decoupling scheme eDUMBO-1_{22} enhances ^{1}H resolution in the indirect dimension by mitigating the effects of residual dipolar coupling, thereby resulting in a well-resolved 2D correlation spectrum at slow MAS. On the other hand, it was also demonstrated that fast MAS significantly improves the resolution and sensitivity of $^{14}N/^{1}H$ correlation spectra using HMQC experiments [21]. At ultra-fast MAS of 90 kHz, ^{14}N heteronuclear dipolar decoupling was implemented to obtain the nitrogen-connected ^{1}H CSA [22]. The application of on-resonance ^{14}N continuous-wave (CW) decoupling during ^{1}H CSA recoupling sequence has been shown to hamper the recoupling of the heteronuclear dipolar coupling, resulting in an undistorted ^{1}H CSA line shape. Recently, Bamola et al. [23] presented an analytical framework which highlights the complexity of spin dynamics of ^{14}N-^{1}H heteronuclear CW decoupling with ^{1}H CSA recoupling under various spinning frequencies. However, experimentally achieving an effective ^{14}N-X spin-spin decoupling remains a challenge at fast MAS.

In this article, we examine the decoupling of ^{14}N interactions to ^{1}H spins in the fast MAS regime using an on-resonance ^{14}N CW irradiation at a static field of 14.1 T. We experimentally show that narrower NH linewidth is obtained using on-resonance low-power decoupling field at a fast MAS rate of 70 kHz. We have also explored the effect of offset, decoupling-field strength, and spinning frequency on the decoupling performance of ^{14}N spin with $C_Q \approx 1.1$–3.4 MHz.

2. Theory

The rotating frame Hamiltonian for a two-spin system (^{1}H and ^{14}N) describing the spin dynamics of ^{14}N CW decoupling is given as follows.

$$H = H_{14N,off} + H_{14N,CS} + H_{14N,rf} + H_Q^{(1)} + H_Q^{(2)} + H_{DD} + H_{RDS} + H_J, \quad (1)$$

where $H_{14N,off}$, $H_{14N,CS}$, $H_Q = H_Q^{(1)} + H_Q^{(2)}$, and $H_{14N,rf}$ represent offset, chemical shift, quadrupolar coupling (first and second-orders), and RF-interaction Hamiltonians for ^{14}N nuclear spin, respectively. The terms H_{DD}, H_{RDS} and H_J represent the ^{14}N-^{1}H heteronuclear-dipolar, RDS, and J couplings, respectively. For simplicity, we have ignored the CSA for both nuclear spins and considered an on-resonance irradiation for the ^{1}H channel. The spin dynamics in Eq. (1) is complicated by the presence of dominating ^{14}N first-order quadrupolar coupling and a non-commuting set of operators corresponding to various interactions. To provide a simplified description of the underlying spin dynamics, Pell et al. [24,25] introduced the description of ^{14}N spin-dynamics in the quadrupolar interaction frame, commonly referred to as the quadrupolar jolting frame (QJF). In this QJF, while the dominating first-order quadrupolar interaction ($H_Q^{(1)}$) is absent, the Hamiltonian corresponding to $H_{14N,off}$, $H_{14N,CS}$, $H_Q^{(2)}$, H_{DD}, H_{RDS} and H_J commute with the first-order quadrupolar coupling Hamiltonian, hence will retain the same form in QJF. However, as described in previous publications [26,27], the RF-interaction Hamiltonian ($H_{14N,rf}$) does not commute with $H_Q^{(1)}$; therefore, the QJF transformation imparts time-dependency to the RF-interaction with a period ($\frac{1}{\nu_Q^{(1)}}$, where $\nu_Q^{(1)}$ is the first-order quadrupolar coupling frequency). Considering irradiation close to the centerband, the average RF-interaction Hamiltonian over a rotor period is given as

$$\overline{H}_{14N,rf} = 2\pi|S_0|\nu_{14N} exp\{-i\phi S_z^2\} S_x \, exp\{i\phi S_z^2\}, \quad (2)$$

where $|S_0| \sim \sqrt{\frac{\nu_r}{\nu_Q^{(1)}}}$ is the RF scaling factor for irradiation at the centerband. It should be noted that explicit $|S_0|$ exhibits orientation dependence and the above relationship only provides a crude order-of-magnitude estimation. Similar equations with a scaling factor of $|S_k|$ hold for irradiation near the kth sideband. Unlike RF-field in the rotating frame (fixed magnitude and orientation (see Fig. 1a)), the strength of the scaled RF-field in QJF will depend on the quadrupolar interaction and the spinning frequency. Additionally, the phase of scaled RF-field depends on crystallite orientations and will be distributed in the X-Y plane (see Fig. 1b). Therefore, in QJF, the spin-dynamics will be equivalent to the spin-1/2 nuclei in the rotating frame, except for the fact that the RF-field is replaced by the scaled RF-field with phase-distribution. Although, different crystallites in the powder sample will experience different magnitudes of the scaled RF-field, but it still acts in transverse direction (X-Y plane), therefore satisfy the necessary requirement for effective decoupling process.

Employing Eq. (2), the QJF Hamiltonian in Eq. (1) can be replaced by MAS averaged first-order Hamiltonian.

$$\overline{H} = 2\pi\Big(-\Omega_{14N} + \nu_{14N,CS} + \overline{\nu}_Q^{(2)}\Big)S_z + 2\pi|S_0|\nu_{14N} \, exp\{-i\phi S_z^2\} S_x \, exp\{i\phi S_z^2\} + 2\pi\overline{\nu}_{RDS} I_z S_z^2 + 2\pi J I_z S_z. \quad (3)$$

Here the ^{14}N offset (Ω_{14N}), ^{14}N chemical-shift ($\nu_{14N,CS}$) and the second-order quadrupolar coupling ($\overline{\nu}_Q^{(2)}$) are combined due to the similar spin-operator dependency for these interactions in $S = 1$ spin systems. $\overline{\nu}_{RDS}$ $\Big(= \nu_{RDS}^{(0)} + \overline{\nu}_{RDS}^{(4)}\Big)$, i.e. both isotropic and 4th rank anisotropic component of RDS) and J (isotropic component) represent the size of RDS and J couplings, respectively. It should be noted that anisotropic interactions are represented by their averaged value under MAS, while heteronuclear dipolar couplings are completely averaged out to the first-order.

Above Eq. demonstrate the decoupling process in QJF, which has a close resemblance to the decoupling process in spin-1/2 system in rotating frame, except that the non-unidirectional transverse RF-field acts in QJF. Fig. 1a-b pictorially highlight the similarities and difference between the decoupling process in rotating and QJF under on-resonance irradiation condition. Subsequently, RF averaging induces transitions in the ^{14}N spin states when on-resonance irradiation: $\Omega_{14N} = \nu_{14N,CS} + \overline{\nu}_Q^{(2)}$ is applied, decoupling RDS and J-couplings. On the other hand, in the presence of offset, the effective RF-field in QJF will not act in transverse plane (X-Y) and would rather be tilted along the Z-axis and the pictorial representation of this distribution is shown in Fig. 1c. Therefore, the effectiveness of the CW decoupling process at faster MAS will be strongly affected by the off-resonance irradiation, unlike the adiabatic decoupling process [19,20]. Thus, we anticipate that the on-resonance CW ^{14}N irradiation would provide the best decoupling efficiency for spin-spin coupling in fast MAS condition. In the following section, we have presented a detailed experimental analysis of the decoupling procedure and optimum decoupling condition.

3. Experiment and sample

All the experimental data were collected using the pulse sequence shown in Fig. 2 on a JEOL spectrometer (Model: JNM-ECZ600R, JEOL Ltd., Japan) equipped with a 1.0 mm double-resonance (HX) fast MAS probe operating at 599.6 MHz (^{1}H) and 43.3 MHz (^{14}N) Larmor frequencies. We have used a naturally abundant N-acetyl L-alanine, L-histidine.HCl.H_2O and tripeptide N-formylmethionyl-leucyl-phenylalanine (fMLF) samples provided by Sigma Aldrich and separately packed in a 1.0 mm zirconia rotor. A spinning frequency of 70 kHz was used, unless otherwise specified. The ^{1}H saturation pulses (N-acetyl L-alanine) and 8 dummy scans (L-histidine.HCl.H_2O and fMLF) were used to stabilize the spin system prior to each scan. The RF power on the ^{14}N nuclei has been calibrated using the uniformly labelled U-^{13}C, ^{15}N, L-alanine sample through the heteronuclear Bloch-Siegert (BS) shift method as described by Gan et al. [28,29]. The on-resonance condition for the ^{14}N

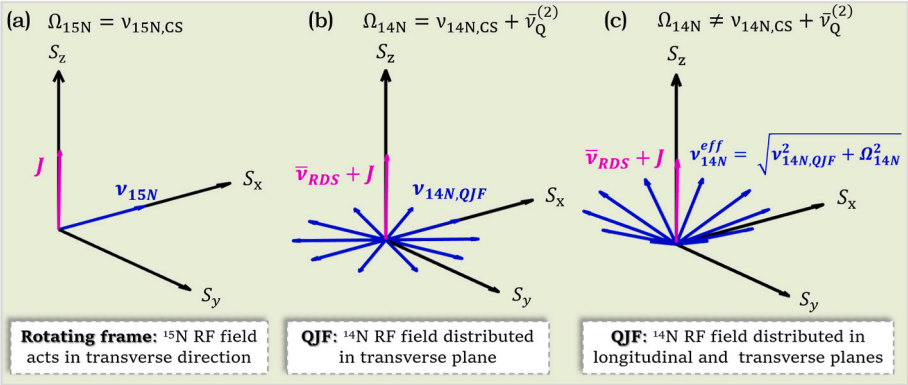


Fig. 1. (a) Decoupling process in the rotating frame in the case of spin-1/2 (e.g., ^{1}H and ^{15}N) systems under on-resonance irradiation at fast MAS. Decoupling process in the QJF in spin-1 and spin-1/2 systems (e.g., ^{1}H and ^{14}N) under ^{14}N on-resonance irradiation (b) and ^{14}N off-resonance irradiation (c) under fast MAS conditions. In this figure, the magenta vector along (Z-axis) represents spin-spin coupling terms, while the blue vectors depict the RF-field acting on the ^{15}N/^{14}N spin. In the rotating frame, the ^{15}N CW RF-field acts in the transverse direction (X) leading to a perfect inversion of the ^{15}N spin states (Z) under on-resonance irradiation $\left(\Omega_{15N} = \nu_{15N,CS}\right)$. In QJF, the scaled RF-field ($\nu_{14N,QJF} = |S_0|\nu_{14N}$) is distributed in the transverse plane (X–Y) under on-resonance irradiation $\left(\Omega_{14N} = \nu_{14N,CS} + \bar{\nu}_Q^{(2)}\right)$. Whereas under off-resonance irradiation $\left(\Omega_{14N} \neq \nu_{14N,CS} + \bar{\nu}_Q^{(2)}\right)$, the effective RF-field acting on the ^{14}N spin in the QJF tilts along the Z-axis leading to distribution in both longitudinal (Z) and transverse (X–Y) planes. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

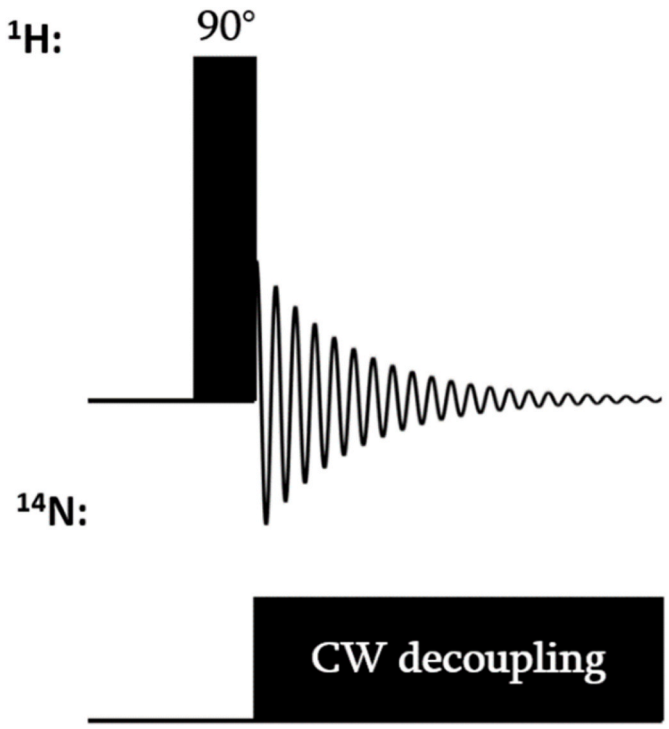


Fig. 2. Schematic representation of the one pulse proton experiment with CW decoupling on the ^{14}N channel.

spin was initially determined through the J based or T-HMQC experiments [13,26] under similar experimental conditions (refer to Fig. S1 in the Supplementary Information for N-acetyl L-alanine), and further optimized experimentally. The chemical shifts of ^{14}N are referenced with respect to nitromethane. The FIDs are Fourier transformed without any line broadening.

4. Results and discussion

Fig. 3 shows chemical structure and resonance assignments of the ^{1}H spectra of N-acetyl L-alanine using the single-pulse experiment (Fig. 2) with and without ^{14}N CW decoupling. In the absence of the ^{14}N decoupling, while the splitting of the ^{1}H resonance at 7.96 ppm is not observed, the comparison of the FWHM of various ^{1}H resonances clearly reveal the underlying broadening of the ^{14}N-connected ^{1}H resonance. The NH peak exhibits a linewidth of 387 Hz, while the OH/CH protons give average linewidth of 192 Hz. We can attribute the observed broadening in NH protons partially to RDS of the ^{1}H directly connected to the ^{14}N. It should be noted that not only RDS but also isotropic J coupling contributes to the line broadening. For comparison, we have also shown the ^{15}N filtered [30] ^{1}H spectrum under ^{15}N WALTZ decoupling of the NH moiety of the same compound at similar experimental conditions. Analyzing resonance widths of ^{1}H directly connected to ^{15}N/^{14}N (i.e. FWHM = 209 Hz for ^{15}N filtered spectra with ^{15}N decoupling, and FWHM = 387 Hz for single-pulse spectrum without ^{14}N decoupling) highlights the presence of the ^{14}N-^{1}H spin-spin coupling and provides an estimate of the maximum achievable line narrowing with efficient ^{14}N decoupling. The implementation of the on-resonance ^{14}N CW decoupling (ν_{14N} = 12.6 kHz, Ω_{14N} = 175 ppm) reduced the line width of the ^{1}H resonance directly connected to ^{14}N (FWHM = 267 Hz), which validates the effectiveness of the ^{14}N-^{1}H decoupling. The small difference in the widths of the ^{1}H{^{14}N} (FWHM = 267 Hz) and ^{1}H{^{15}N} (FWHM = 209 Hz) decoupled spectra could be attributed to the different magnitudes of the BS shifts and residual RDS coupling in the former case. In the next section, we will discuss the effect of the BS shifts and associated broadening as well as the origin of the residual broadening observed.

Even though the implementation of the ^{14}N CW decoupling scheme is straightforward, achieving an optimum decoupling efficiency necessitates the meticulous optimization of three experimental parameters: decoupling field strength (ν_{14N}), offset (Ω_{14N}), and spinning frequency (ν_r). To evaluate the effect of ^{14}N RF-field strength on the decoupling efficiency, we carried out the experiments by varying the ^{14}N RF-field amplitudes from 0 to 42 kHz under the on-resonance irradiation condition (Ω_{14N} = 175 ppm), as depicted in Fig. 4. The variation in the spectral intensity of NH resonance (around 7.96 ppm) is inversely related to the variation in linewidth under the effect of CW decoupling, while maintaining constant area integral. Our observation indicates that moderate decoupling fields, specifically those between ν_{14N} = 8.4–16.8 kHz provide more effective decoupling as compared to higher field strengths. While a stronger decoupling field may result in better decoupling, it can also introduce significant peak shifts and broadenings

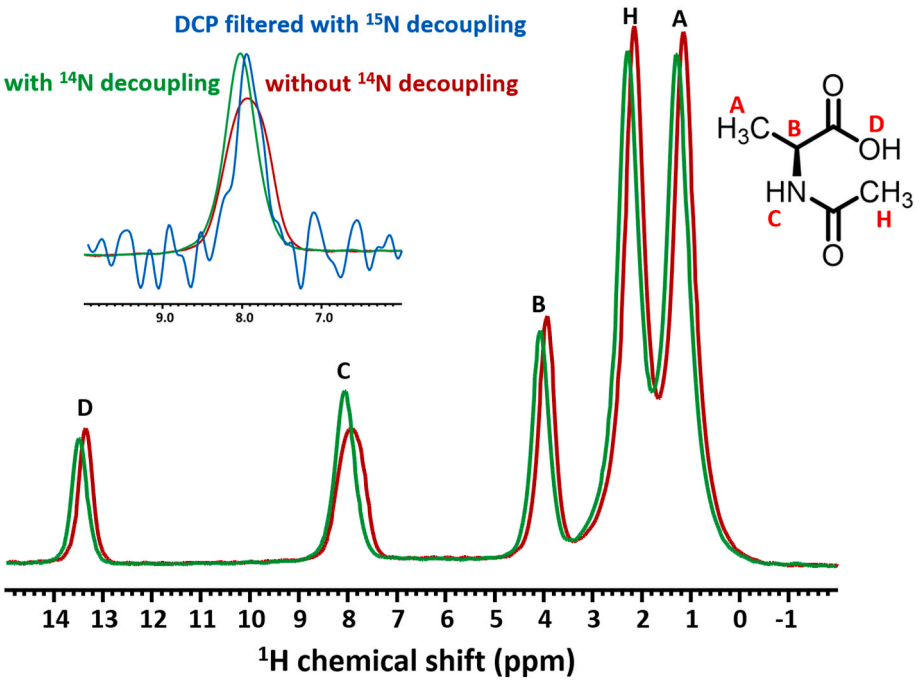


Fig. 3. Chemical structure and chemical shift assignments of the ^{1}H spectra of N-acetyl L-alanine with (green) and without (red) ^{14}N decoupling at a spinning frequency of 70 kHz. In the green curve, the optimal CW decoupling condition ($\nu_{14N} = 12.6$ kHz and $\Omega_{14N} = 175$ ppm) is used. The inset represents the comparison of the ^{14}N decoupled ^{1}H peak with the ^{15}N DCP filtered ^{1}H signal under ^{15}N WALTZ decoupling (blue curve). The NH line widths in ^{1}H spectra without ^{14}N decoupling, with ^{14}N decoupling and DCP filtered ^{15}N decoupling are 387 Hz, 267 Hz and 209 Hz, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

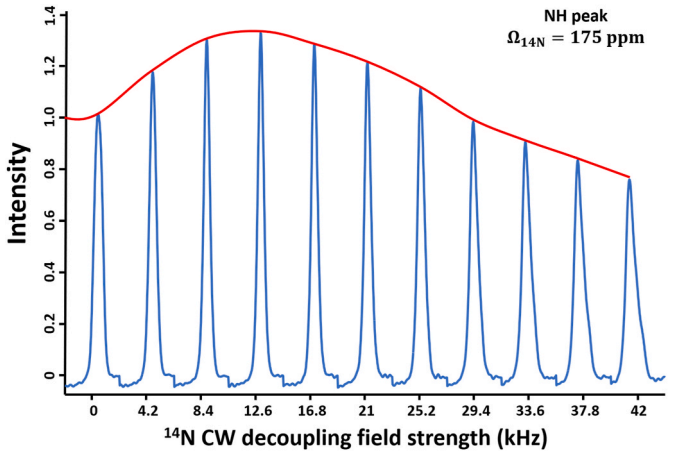


Fig. 4. The intensity variation of the NH peak for N-acetyl L-alanine at various ^{14}N decoupling field strengths under on-resonance ^{14}N irradiation ($\Omega_{14N} =$ 175 ppm). The spectral intensity is normalized with respect to the non-decoupled ^{1}H intensity.

due to the Bloch-Siegert (BS) effect, especially in the presence of B_1 field inhomogeneities.

Here, we will discuss the effect of BS shift [28,29]. The heteronuclear BS shifts on ^{1}H under X CW irradiation can be calculated by

$$\nu_{cal}^{BS}\,(in\ Hz) = \frac{\gamma_{1H}^2}{\gamma_X^2}\,\frac{\nu_0^{1H}\nu_X^2}{\left(\nu_0^{1H}\right)^2 - \left(\nu_0^X\right)^2} \tag{4}$$

where $\gamma_{1H/X}$ and $\nu_0^{1H/X}$ represent gyromagnetic ratio and Larmor precession frequency for ^{1}H/X nucleus and $\nu_X = \gamma_X B_1$ is the RF-field acting on X spin. The effect of BS shift under various ^{14}N decoupling irradiation is substantiated in Fig. 5. Fig. 5a shows comparison between experimentally observed variation in the NH resonance and calculated BS shift, combined with B_1 inhomogeneity, under various ^{14}N decoupling field strengths. Fig. 5b shows a good correlation between the calculated BS shift and the experimentally observed peak shifts across decoupling field strength in the range 0–42 kHz. This correlation justifies the presence of the heteronuclear BS phenomenon under ^{14}N-^{1}H CW decoupling, which induces a shift in the overall ^{1}H spectrum (Fig. S3 in the Supplementary Information) under decoupling. The observed experimental shift in the overall ^{1}H spectra under ^{14}N ($\nu_{14N} = 12.6$ kHz) and ^{15}N (inset at $\nu_{15N} = 10$ kHz) decoupling in Fig. 3 were calculated to be 51 and 16.2 Hz, respectively. To achieve a precise estimation of ^{1}H isotropic chemical shifts under continuous decoupling irradiations, it is always important to take care of reference to include heteronuclear BS shift.

BS shift introduces not only shift in peak position but also spectral broadening when B_1 inhomogeneity presents. The calculated spectral broadening under BS effect, considering a hypothetical +10% to −15% asymmetric distribution of B_1 field (error bar in blue) is shown in Fig. 5a and agree with the experimentally observed broadening [31]. To further support the BS induced spectral broadening under ^{14}N decoupling, a comparison between the line widths of NH and OH versus the calculated BS shifts at ^{14}N decoupling field strengths is presented in Fig. 5c. The additional line width of the OH proton is predominantly influenced by the BS shifts induced under the ^{14}N decoupling irradiation and this is clear by the linear slope of OH line widths at varied the calculated BS shift. In addition, the asymmetric broadening of OH and NH resonances towards low-frequency region (Fig. S3 in the Supplementary Information) provides direct evidence for spectral broadening induced by asymmetric inhomogeneous B_1 field. It is important to note here is that the line widths in Fig. 5c are plotted at 20% of the peak intensities instead of FWHM. As FWHM is dominated by the natural line widths, the nonlinear behavior of the FWHM (OH proton) versus heteronuclear BS shifts is observed as shown in Fig. S2. Note that the heteronuclear BS shifts/broadenings on ^{1}H spectra (high γ-nuclei) under continuous decoupling irradiation on low γ-nuclei are unavoidable. For instance, in

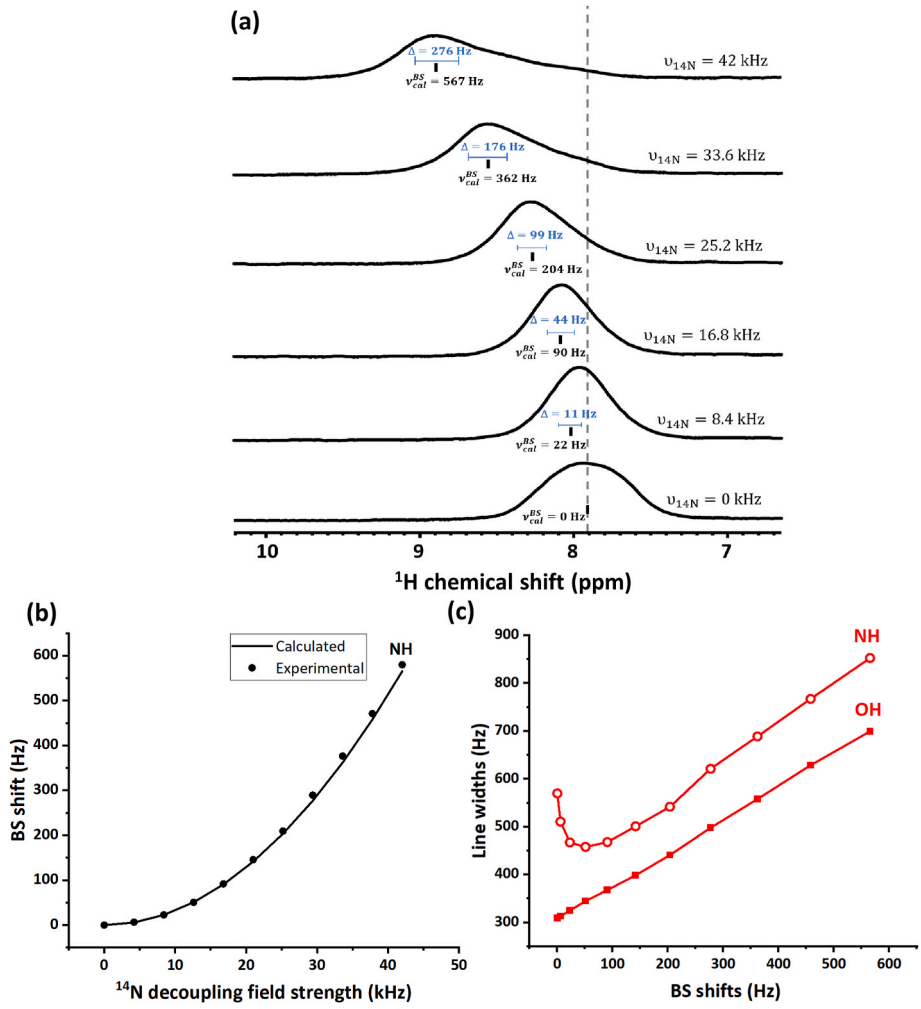


Fig. 5. (a) Comparison of the calculated BS shifts (ν_{cal}^{BS}) and the experimentally observed shift in the NH resonance at different ^{14}N decoupling field strengths (as mentioned in the figure) under on-resonance irradiation conditions (Ω_{14N} = 175 ppm) for N-acetyl L-alanine. The error bar (Δ) in blue color represents the effect of +10% to −15% of B_1 inhomogeneities in the calculated BS values (ν_{cal}^{BS}). (b) Variation of the calculated (black curve) and experimental (black solid circle) BS shifts as a function of ^{14}N decoupling field in the range from 0 to 42 kHz. (c) Variation of the experimental NH (red curve with open circle) and OH (red curve with closed square) line widths at 20% of maximum spectral intensity versus calculated BS shifts at different of ^{14}N decoupling field strengths in the range from 0 to 42 kHz. Solid lines in panel c are guide for eye. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Fig. S4 of the Supplementary Information, the effect of ^{13}C decoupling under three different strengths (ν_{13C} = 0, 52 and 104 kHz) at Ω_{13C} = 100 ppm at fast MAS of 70 kHz is shown.

On the other hand, the plot of the NH proton line widths versus calculated BS shifts experiences a minimum and then a subsequent increase under the ^{14}N decoupling field variation in the range of 0–42 kHz. The observed plot of the NH linewidth as a function of ^{14}N decoupling field strength results from two competing phenomenon: line narrowing due to ^{14}N decoupling and line broadening caused by BS shifts under B_1 inhomogeneities. At lower decoupling field strength, the magnitude of the BS shifts is relatively small, therefore broadening induced by B_1 inhomogeneity is insignificant compared to natural NH linewidth. With this condition, the effect of ^{14}N-^{1}H decoupling is clearly visible as a reduced NH linewidth. Whereas at higher ^{14}N RF-field strength (ν_{14N} = 30–42 kHz, Fig. 5c), the NH line width is predominantly described by the BS broadening over the decoupling effect revealed by nearly linear slope of calculated BS shifts and NH widths. It should be noted that the slope of the NH peak is almost identical to that of the OH peak, corroborating the dominant broadening due to the BS effect. Thus, the low-power ^{14}N irradiation appears to be the best compromise towards improved resolution of ^{1}H sites coupled to ^{14}N spins at fast MAS.

It is important to note that the observed broadening of NH linewidth at stronger decoupling irradiations can be partially attributed to the TRAPDOR recoupling phenomenon as commonly observed in the quadrupolar connected spin-1/2 nuclei. Delevoye et al. [17,18] explained that TRAPDOR recoupling is responsible for increased ^{31}P {^{27}Al} linewidth under CW at slow MAS. TRAPDOR utilize the adiabatic energy mixing at stronger RF-fields and slow MAS. As the adiabatic modulation of ^{14}N spin states does not hold at faster MAS and decoupling strength used in this paper, therefore it negates the presence of TRAPDOR recoupling. This is evident from the nearly similar tilt of NH and OH resonances. This argument is further strengthened by the observed strong off-resonance dependence of the ^{14}N-^{1}H decoupling effect as discussed later. To verify the negligible contribution of TRAPDOR recoupling to NH linewidths, we have plotted the NH and OH spectral intensities obtained in the ^{1}H spin-echo experiment with ^{14}N decoupling during constant echo duration. No decoupling was implemented during detection period. The pulse sequence and experimental details for this implementation are shown in Fig. S5 in the Supplementary Information. Evolution under ^{1}H chemical shift, BS shift, RDS and J couplings are refocused by the 180° pulse, therefore this implementation is a good choice for studying the TRAPDOR induced recoupling phenomenon without any line broadening effect of BS phenomenon. The experimental data were acquired under two different phase settings of

the second decoupling pulse. In one experiment, the same phases of ^{14}N decoupling were used before ($\phi_1 = 0°$) and after ($\phi_2 = 0°$) the 180° pulse, while the other adopted $\phi_1 = 0°$, $\phi_2 = 180°$. We performed the experiment under on-resonance condition ($\Omega_{14N} =$ 175 ppm) mentioned in this paper and off-resonance condition at first spinning sideband which maximizes the TRAPDOR recoupling ($\Omega_{14N} = \nu_r =$ 70 kHz). The OH resonance exhibits a similar spectral intensity across all decoupling strengths, suggesting no TRAPDOR effect [refer to Fig. S6a2-b2]. However, the NH resonance shows reduced intensities with increasing decoupling field strength through TRAPDOR effect. It is interesting that its behaviour differs under the two-phase settings [refer to Fig. S6a1]. Nevertheless, the maximum TRAPDOR decay was observed for $\phi_1 = 0°$ and $\phi_2 = 180°$. The decay yields T_2 of 19.7 ms and 12.3 ms under on-resonance irradiations at $\nu_{14N} = 12.7$ kHz and 41.3 kHz, corresponding to the NH linewidths contributions of 16 Hz and 26 Hz, respectively. These observations indicate a minimal TRAPDOR contribution to NH linewidths, even under strong decoupling irradiation. As shown before, larger TRAPDOR effect is observed when RF-field is applied to maximized at a spinning sideband ($\Omega_{14N} = \nu_r =$ 70 kHz) (Fig. S6b1) [26]. In this case, the T_2 decay constant was observed to be 30 ms and 8.3 ms using $\nu_{14N} = 12.7$ kHz and 41.3 kHz, which yields 10.5 Hz and 38 Hz contributions to the NH linewidth, respectively. Since the low power decoupling strategies use $\nu_{14N} = 12.7$ kHz, therefore, above observations confirm the negligible TRAPDOR induced dipolar recoupling contributing to the NH linewidths under irradiation near both centerband and spinning sideband. It is important to note here that the slightly steeper tilt of the NH proton than that of OH proton in Fig. 5c may be related to the TRAPDOR recoupling at stronger decoupling irradiation. Thus, the above observations can be concluded as that the low-power ^{14}N irradiation results in efficient spin-spin decoupling at fast MAS. From an experimental standpoint, the low-power decoupling produces minimal power deposition and heating, hence reducing the risk of probe damage and increasing the maximum decoupling time.

A serious limitation of low-power decoupling is its sensitivity to the off-resonance effects. In order to investigate the effect of the off-resonance irradiations (Ω_{14N}) on decoupling efficiency, we performed the experiment with varying ^{14}N offsets ranging from −400 ppm to 400 ppm, and the decoupling trends are shown for the NH moiety in Fig. 6a. We observed an overall decrease in the width of the ^1H resonance (directly related to NH signal intensity) within the ^{14}N offset range of 100–300 ppm, which lies in the expected range of ^{14}N resonance at 175 ppm (refer to Fig. S1 in the Supplementary Information). The theoretical explanation of this phenomenon is provided in the theory section based on the tilt of the effective RF-field in QJF under ^{14}N off-resonance irradiation. Note that, in the powder sample, due to the orientation-dependent isotropic second-order quadrupolar shifts, it is not possible to be exact on-resonance for all the crystallite orientations simultaneously, resulting in an inherent compromise in the ^1H-^{14}N decoupling efficiency. This factor could possibly have a bearing on the residual width of the amide proton under on-resonance ^{14}N decoupling (FWHM = 267 Hz) in comparison to the ^{15}N-filtered ^1H DCP spectrum (FWHM = 209 Hz) (Fig. 3). A part of the broadening can be ascribed to the different BS effect in $^1H\{^{14}N\}$ and $^1H\{^{15}N\}$ decoupling, mentioned previously.

Fig. 6b,c demonstrates the effect of irradiation near the spinning sideband positions ($\Omega_{14N} = \pm\nu_r$). We observe a similar trend of the decoupling efficiencies (Fig. 6a) for irradiation at the spinning sideband positions, with the maximum decoupling efficiency occurring at $\Omega_{14N} - \nu_r = 150$ ppm and $\Omega_{14N} + \nu_r = 175$ ppm. A comparison between the

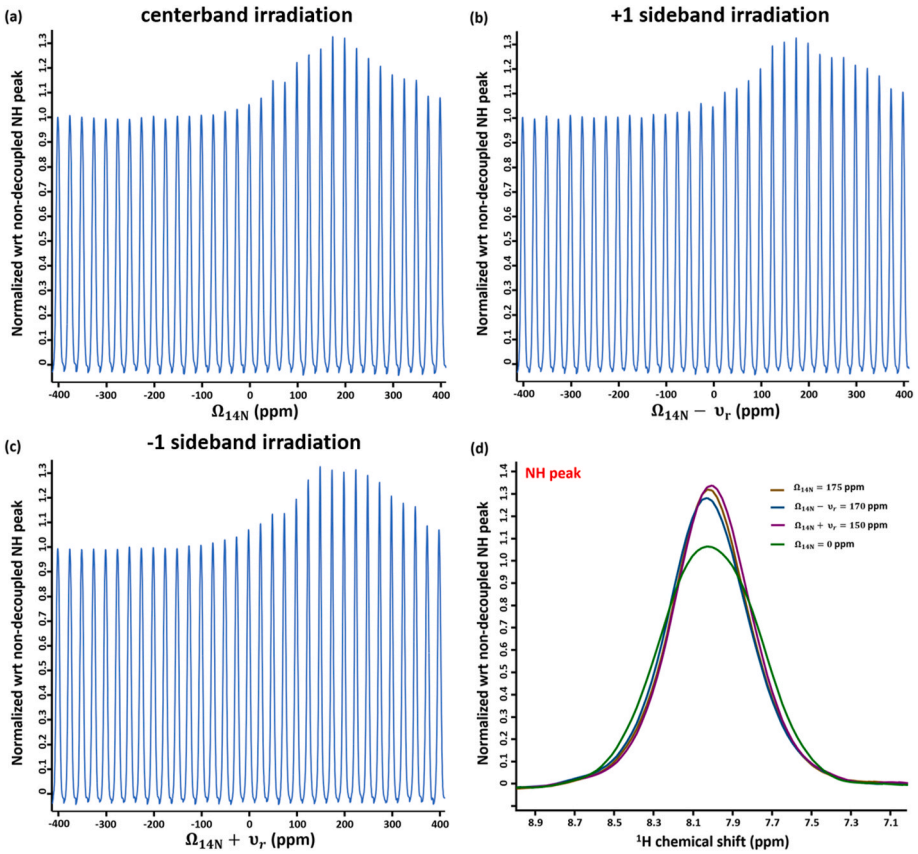


Fig. 6. (a–c) Variation of the spectral intensity of the NH resonance for N-acetyl L-alanine versus the ^{14}N offsets at a decoupling field strength of $\nu_{14N} =$ 12.6 kHz and $\nu_r = 70$ kHz. The variation of the ^{14}N offset (with the step size of 25 ppm) is presented near (a) centerband, (b)+1 sideband and (c) −1 sideband conditions. The spectral intensity of NH resonance is normalized with respected to the non-decoupled ^1H signal intensity. (d) Comparison of the ^{14}N-^1H decoupling efficiency at respective optimized ^{14}N offsets in panels a–c with $\Omega_{14N} = 0$ ppm at the specified decoupling irradiation strength.

various off-resonance irradiation conditions is shown in Fig. 6d, where the irradiations at $\Omega_{14N} = 0$ ppm, $\Omega_{14N} = 175$ ppm, $\Omega_{14N} - \nu_r = 150$ ppm and $\Omega_{14N} + \nu_r = 175$ ppm result in nearly equivalent ^{1}H signal intensity. The nearly identical decoupling dynamics at centerband and sideband irradiation, is due to minimal contribution of the TRAPDOR induced broadening at low-power decoupling condition as discussed above. As shown above, it is important to emphasize that the ^{14}N decoupling is highly sensitive to the off-resonance irradiations; thus, a shift of 50 ppm from the centerband or sideband results in 6% drop in spectral intensities. The observed offset sensitivity of the decoupling effect further validates the proposed decoupling mechanism which contrast with the existing adiabatic ^{14}N decoupling sequence at slow MAS, which are nearly insensitive to ^{14}N off-resonance irradiation [19, 20].

To further demonstrates the practical utility and universality of the proposed CW decoupling process, we have shown a comparison between the ^{14}N decoupled and non-decoupled T-HMQC filtered ^{1}H spectra for L-histidine.HCl.H_2O ($C_Q \approx 1.14$–1.46 MHz) and a tripeptide N-formylmethionyl-leucyl-phenylalanine (fMLF) ($C_Q \approx 2.7$–3.0 MHz) samples in Fig. 7. While optimizing the decoupling field strengths, we observed a similar decoupling trend as presented in Figs. 4 and 5 (data not shown). At low-power on-resonance ^{14}N irradiation ($\nu_{14N} =$ 6.3 and 10.2 kHz for L-histidine.HCl.H_2O and fMLF, respectively) provided the narrowest NH linewidths. The improved resolution for overlapping NH resonances in fMLF emphasize the importance of simple ^{14}N-^{1}H CW decoupling in sample containing multiple nitrogen sites. The additional benefit of improved resolution is an enhancement in spectral intensity under decoupling, leading to a reduction in experimental time. For instance, a 1.3-fold increase in signal intensity for fMLF under ^{14}N decoupling yields a 1.7-fold decrease in experimental time to achieve a comparable S/N ratio. On the other hand, systems with smaller C_Q's like L-histidine.HCl.H_2O exhibits limited resolution enhancement due to inherent small RDS and large inhomogeneous broadening.

To further illustrate the role of spinning frequency (ν_r), we conducted decoupling experiments at variable MAS rates of 20, 40, and 70 kHz under their respective optimized decoupling conditions mentioned in Fig. 8 caption for N-acetyl L-alanine. A comparison of the ^{1}H spectra with and without ^{14}N-^{1}H decoupling at various spinning speeds highlights the necessity of faster MAS for improved ^{1}H signal sensitivity and better decoupling efficiency. At slower MAS (40 and 20 kHz), ^{14}N irradiation does a little to no improvement in the NH signal intensity. Fig. S7 in the Supplementary Information shows NH spectral intensities at different spinning frequencies (20, 40 and 70 kHz) plotted as a function of ^{14}N decoupling field strength under on-resonance ^{14}N irradiation. The inefficient low-power decoupling performance may be attributed to the small magnitude of scaled RF in the QJF ($\propto\sqrt{\nu_r}$), which is insufficient for manipulating ^{14}N spin states. While this limitation may be overcome with stronger decoupling irradiation, the NH linewidths become broader due to amalgamation of BS shifts/broadenings and TRAPDOR recoupling. In contrast, at faster MAS (70 kHz), low-power RF averaging becomes significantly effective due to relatively larger magnitude of the scaled RF-field and reduced homogeneous linewidth, leading to improved decoupling performance. Thus, the combination of fast MAS and low-power on-resonance ^{14}N CW irradiation is an essential and optimal condition for effective ^{14}N-^{1}H spin-spin decoupling. In princple, ^{14}N decoupling at overtone frequency can also be used to influece the ^{1}H

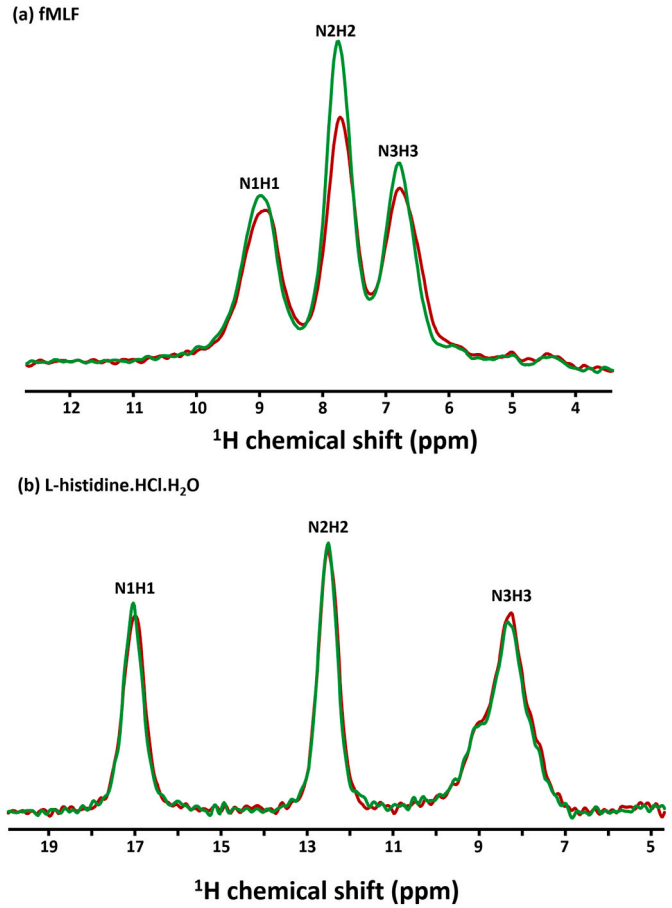


Fig. 7. Comparison of ^{14}N T-HMQC filtered non-decoupled and CW decoupled {^{14}N}^{1}H NMR spectra for tripeptide N-formyl Met-Lue-Phe (fMLF) (a) and L-histidine.HCl.H_2O (b) at fast MAS of 70 kHz. Decoupling experiments were performed using the optimized decoupling condition $\nu_{14N} = 10.2$ kHz at $\Omega_{14N} =$ 120 ppm and $\nu_{14N} = 6.3$ kHz at $\Omega_{14N} =$ -110 ppm for fMLF and L-histidine.HCl.H_2O, respectively. FWHM for non-decoupled (decoupled) NH resonances are as follow: (a) N1-H1: 441.3 Hz (395 Hz), N2-H2: 325 Hz (250 Hz) and N3-H3: 424 Hz (348 Hz) and (b) N1-H1: 327.6 Hz (318 Hz), N2-H2: 328Hz (289 Hz) and N3-H3: 531 Hz (518 Hz).

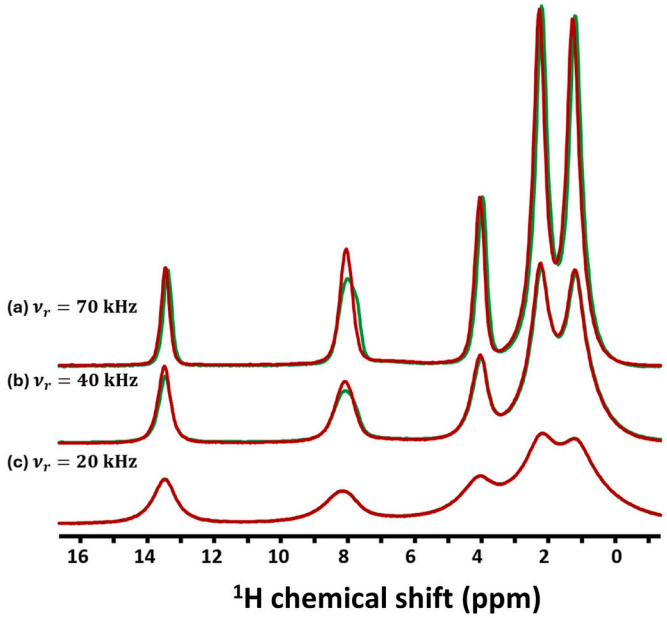


Fig. 8. Effect of the three different spinning frequencies $\nu_r = (a)70, (b)40$ and (c) 20 kHz on the ^{14}N-^{1}H decoupling efficiency for N-acetyl L-alanine at their respective optimized conditions. Here, the red and green curves represent ^{1}H spectra with ($\nu_{14N} = 12.6$ (a), 9.14 (b) and 0 kHz at on-resonance irradiations $\Omega_{14N} = 175$ ppm) and without ^{14}N CW irradiations, respectively. FWHM for NH resonances in non-decoupled (decoupled) spectra are 408.3, 478.6 and 722 Hz (267.68, 413.7 and 722 Hz) under MAS rates 70 (a), 40 (b) and 20 kHz (c), respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

linewidth and was previously demostrated in the context of ^{13}C spectroscopy [32].

5. Conclusion

We have presented decoupling of ^{14}N-^{1}H spin-spin interactions under ^{1}H detection. Experimental findings show that employing on-resonance low-power CW ^{14}N irradiation provides narrow linewidth at a MAS rate of 70 kHz. ^{14}N decoupling reduces FWHM of NH proton in N-acetyl L-alanine from 387 Hz to 267 Hz and in tripeptide fMLF from 325 Hz to 250 Hz (N2-H2), leading to 25–30% sensitivity enhancement in both cases. As the decoupling is sensitive to off-resonance irradiation, the decoupling will be compromised in the sample with the multiple ^{14}N sites with large chemical shift spread where the condition of simultaneous on-resonance irradiation will not be satisfied for all ^{14}N nuclei. This ^{14}N decoupling condition is a result of careful optimization of RF-field, ^{14}N offsets, MAS frequency and BS induced broadening. We are currently investigating more robust ^{14}N decoupling sequences in our laboratory.

Declaration of competing interest

Ekta Nehra and Yusuke Nishiyama are employees in JEOL Ltd.

Acknowledgement

This work is supported by Department of Atomic Energy (DAE), Government of India (GOI), under Project Identification No. RTI 4007. VA would like to acknowledge the TIFR National Facility of High Field NMR at Hyderabad.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ssnmr.2025.102006.

Data availability

Data will be made available on request.

References

[1] T. Le Marchand, T. Schubeis, M. Bonaccorsi, P. Paluch, D. Lalli, A.J. Pell, L. B. Andreas, K. Jaudzems, J. Stanek, G. Pintacuda, 1H-Detected biomolecular NMR under fast magic-angle spinning, Chem. Rev. 122 (10) (2022) 9943–10018.
[2] S. Ahlawat, K.R. Mote, N.A. Lakomek, V. Agarwal, Solid-state NMR: methods for biological solids, Chem. Rev. 122 (10) (2022) 9643–9737.
[3] Y. Nishiyama, G. Hou, V. Agarwal, Y. Su, A. Ramamoorthy, Ultrafast magic angle spinning solid-state NMR spectroscopy: advances in methodology and applications, Chem. Rev. 123 (3) (2023) 918–988.
[4] Y. Nishiyama, Editorial – ultrafast MAS NMR, J. Magn. Reson. 354 (2023) 107525.
[5] Y. Nishiyama, Fast magic-angle sample spinning solid-state NMR at 60–100 kHz for natural abundance samples, Solid State Nucl. Magn. Reson. 78 (2016) 24–36.
[6] M.M. Maricq, J.S. Waugh, NMR in rotating solids, J. Chem. Phys. 70 (7) (1979) 3300–3316.
[7] E.R. Andrew, A. Bradbury, R.G. Eades, Nuclear magnetic resonance spectra from a crystal rotated at high speed, Nature 182 (4650) (1958) 1659.
[8] Michael H. Frey, S.J. Opella, High-resolution features of the 13C N.M.R. Spectra of solid amino acids and peptides, J. Chem. Soc., Chem. Commun. (11) (1980) 474–475.
[9] C.J. Groombridge, R.K. Harris, K.J. Packer, B.J. Say, S.F. Tanner, High-resolution 13C N.M.R. spectra of solid nitrogen-containing compounds, J. Chem. Soc., Chem. Commun. (4) (1980) 174–175.
[10] A. Llor, J. Virlet, Towards high-resolution NMR of more nuclei in solids: sample spinning with time-dependent spinner Axis angle, Chem. Phys. Lett. 152 (2,3) (1988) 248–253.
[11] A. Samoson, E. Lippmaa, Synchronized double-rotation NMR spectroscopy, J. Magn. Reson. 84 (1989) 410–416.
[12] S. Cavadini, S. Antonijevic, A. Lupulescu, G. Bodenhausen, Indirect detection of nitrogen-14 in solids via protons by nuclear magnetic resonance spectroscopy, J. Magn. Reson. 182 (1) (2006) 168–172.
[13] S. Cavadini, A. Lupulescu, S. Antonijevic, G. Bodenhausen, Nitrogen-14 NMR spectroscopy using residual dipolar splittings in solids, J. Am. Chem. Soc. 128 (24) (2006) 7706–7707.
[14] Z. Gan, Measuring amide nitrogen quadrupolar coupling by high-resolution 14N/13C NMR correlation under magic-angle spinning, J. Am. Chem. Soc. 128 (18) (2006) 6040–6041.
[15] A. Naito, S. Ganapathy, K. Akasaka, C.A. McDowell, Chemical shielding tensor and 13C-14N dipolar splitting in single crystals of L-alanine, J. Chem. Phys. 74 (1981) 3190–3197.
[16] A.C. Olivieri, L. Frydman, M. Grasselli, L.E. Diaz, Analysis of 13C, 14N residual dipolar coupling in the 13C CP/MAS NMR spectra of ribonucleosides, Magn. Reson. Chem. 26 (4) (1988) 281–286.
[17] L. Delevoye, C. Fernandez, C.M. Morais, J-P. Amoureux, V. Montouillout, J. Rocha, Double-resonance decoupling for resolution enhancement of 31P solid-state MAS and 27Al → 31 P MQHETCOR NMR, Solid State Nucl. Magn. Reson. 22 (4) (2002) 501–512.
[18] L. Delevoye, J. Trébosc, Z. Gan, L. Montagne, J-P. Amoureux, Resolution enhancement using a new multiple-pulse decoupling sequence for quadrupolar nuclei, J. Magn. Reson. 186 (1) (2007) 94–99.
[19] R.M. Orr, M.J. Duer, Decoupling residual dipolar coupling between 13C and14N spin pairs in CPMAS NMR, Solid State Nucl. Magn. Reson. 30 (3–4) (2006) 130–134.
[20] R.S. Stein, B. Elena, L. Emsley, Improving resolution in proton solid-state NMR by removing nitrogen-14 residual dipolar broadening, Chem. Phys. Lett. 458 (4–6) (2008) 391–395.
[21] Y. Nishiyama, Y. Endo, T. Nemoto, H. Utsumi, K. Yamauchi, K. Hioka, T. Asakura, Very fast magic angle spinning 1H-14N 2D solid-state NMR: sub-micro-liter sample data collection in a few minutes, J. Magn. Reson. 208 (1) (2011) 44–48.
[22] M.K. Pandey, Y. Nishiyama, Determination of NH proton chemical shift anisotropy with 14N-1H heteronuclear decoupling using ultrafast magic angle spinning solid-state NMR, J. Magn. Reson. 261 (2015) 133–140.
[23] N. Bamola, M. Bansal, R. Ramachandran, Quantifying quadrupole effects in the NMR spectra of spin-1/2 nuclei in rotating solids, Phys. Chem. Chem. Phys. 25 (27) (2023) 17877–17900.
[24] A.J. Pell, K.J. Sanders, S. Wegner, G. Pintacuda, C.P. Grey, Low-power broadband solid-state MAS NMR of 14N, J. Chem. Phys. 146 (2017) 194202.
[25] P. Caravatti, G. Bodenhausen, R.R. Ernst, Selective Pulse Experiments in High-Resolution Solid State NMR, J. Magn. Reson. 55 (1) (1983) 88–103.
[26] I. Hung, P. Gor'kov, Z. Gan, Efficient and sideband-free 1H-detected 14N magic-angle spinning NMR, J. Chem. Phys. 151 (15) (2019) 154202.
[27] V. Agarwal, S. Raran-Kurussi, Y. Nishiyama, Spin-dynamics and efficiency of single 14N-1H cross-polarization at fast magic angle spinning in solids, Solid State Nucl. Magn. Reson. 136 (2025) 101992.
[28] I. Hung, P. Gor'kov, Z. Gan, Using the heteronuclear Bloch-Siegert shift of protons for B1 calibration of insensitive nuclei not present in the sample, J. Magn. Reson. 310 (2020) 106636.
[29] F. Bloch, A. Siegert, Magnetic resonance for nonrotating fields, Phys. Rev. 57 (1940) 522–527.
[30] Y. Ishii, R. Tycko, Sensitivity enhancement in solid state 15N NMR by indirect detection with high-speed magic angle spinning, J. Magn. Reson. 142 (1) (2000) 199–204.
[31] K. Takeda, Microcoils and microsamples in solid-state NMR, Solid State Nucl. Magn. Reson. 47–48 (2012) 1–9.
[32] J. Fukazawa, S.I. Kato, T. Ozaki, A. Shoji, K. Takegoshi, 14N quadrupolar coupling of amide nitrogen and peptide secondary structure as studied by solid-state NMR spectroscopy, J. Am. Chem. Soc. 132 (12) (2010) 4290–4294.